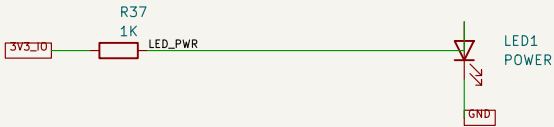
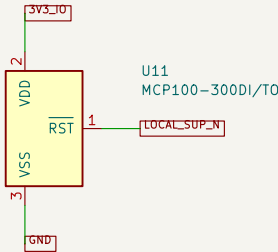


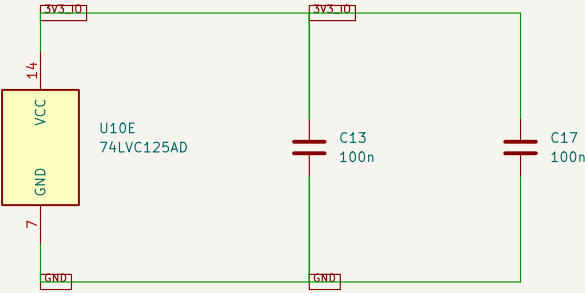
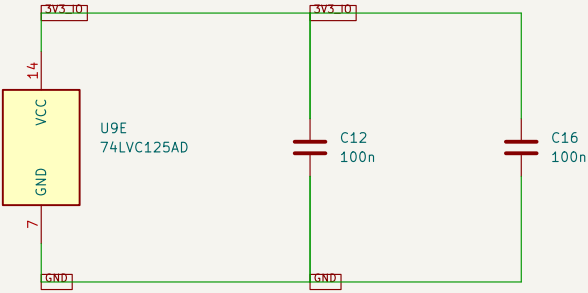
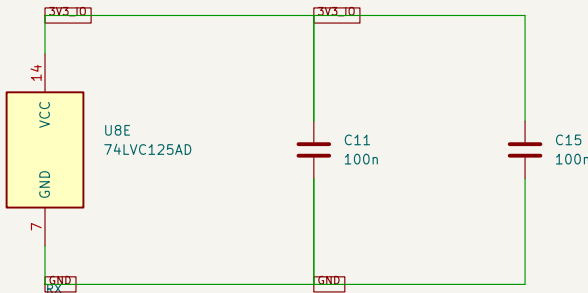
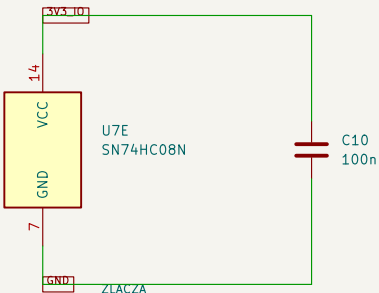
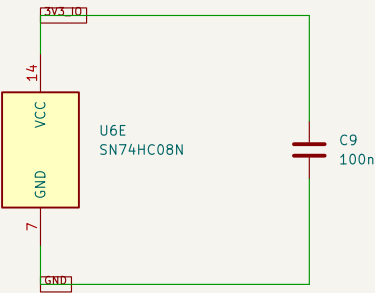
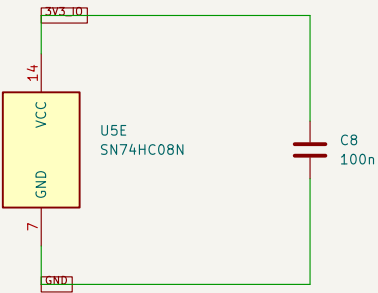
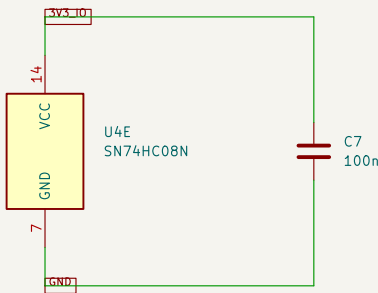
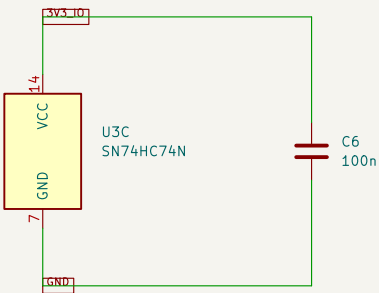
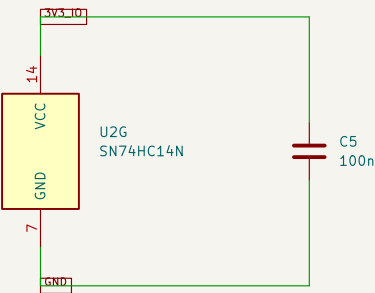
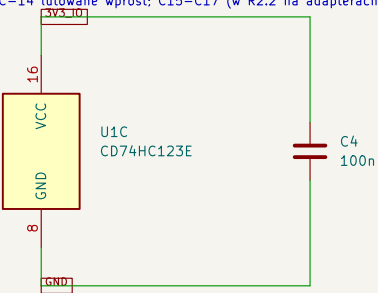
#FLG1
PWR_FLAG

#FLG2
PWR_FLAG



Zasilanie 3V3_IO z P02 przez P12 (J_BP2.10, J_BP3.10). 5V_SYS (J_BP3.8) idzie tylko na kolek J_SV3.2; nie zasila logiki.

R3: U8-U10 SOIC-14 lutowane wprost; C15-C17 (w R2.2 na adapterach) jako druga 100n przy pinach 14/7 U8-U10.



Plik: ZLACZA.kicad_sch

ILK

Plik: ILK.kicad_sch

OUT

Plik: OUT.kicad_sch

Plik: RX.kicad_sch

WD

Plik: WD.kicad_sch

SERWIS

Plik: SERWIS.kicad_sch

U11: MCP100 w bondout D, 1 RESET / 2 VDD / 3 GND. Lokalny reset nie jest podłączony push-pull do SAFE_N.

Hardware interlock, watchdog and manual arm; format S1 (L, level 6); schematic only, hardware not tested

Prototype / schematic + PCB review

Sheet: /

File: P04.kicad_sch

Title: EGRLab P04 SAFE

Size: A3 Date: 2026-10-05

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Id: 1/7

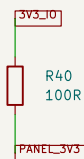
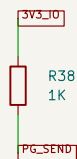
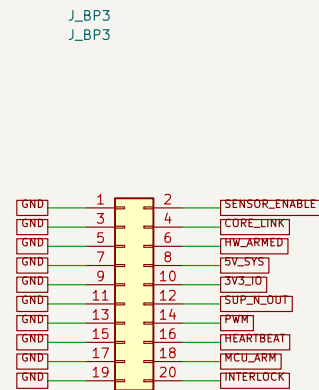
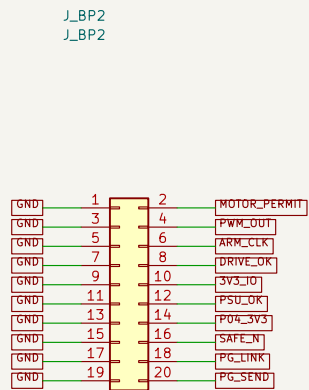
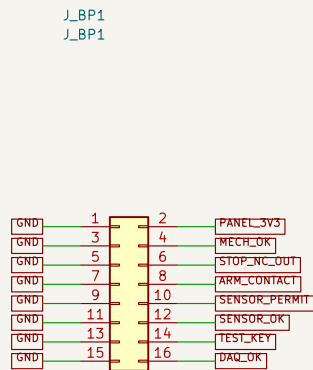
P04-R3 / 02	ZLACZA J_BP1-J_BP3 DO P12 (KRAWEDZ A)
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R3 (S1 5): katowe obudowane IDC na krawedzi A, po jednym w slocie: J_BP1 S1 (x=26.5, 2x8), J_BP2 S2 (x=80.0, 2x10), J_BP3 S3 (x=133.5, 2x10).

Pin 1 od strony mniejszego x, wszystkie nieparzyste GND. Tasma ok. 30 mm do P12; P12 łączy sieci po nazwie. Zastępują J1–J8 z R2.2.

J_BP1: panel P11 (PANELSAFE, piny 2/4/6/8/14 jak J_P12), SENSOR do P08, DAQ_OK z P05. J_BP2: DRIVE do P07 (czeka na P07), P02 (12/16/18 jak J_BP P02 R4).

J_BP3: P03 R6 (12/14/16/18/20 jak J_BP3 P03), 5V_SYS tylko na kolek serwisowy, 3V3_IO zasila plytke (takze J_BP2.10).



3V3 wychodzi z płytki tylko przez rezystory: PANEL_3V3 R40 100R (J_BP1.2), P04_3V3 R39 1k (J_BP2.14), PG_SEND R38 1k (J_BP2.20).

SAFE_N (J_BP2.16): wspólny węzeł z otwartymi kolektorami; P12 łączy go z P02 R4 (Q7) i P07. Nazwy SUP_N_OUT i P04_3V3 jak w P12 (w R2.2: SUP_N, PG_3V3).

Hardware interlock, watchdog and manual arm; format S1 (L, level 6); schematic only, hardware not t

Prototype / schematic + PCB review

Sheet: /ZLACZA/
File: ZLACZA.kicad_sch

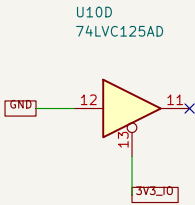
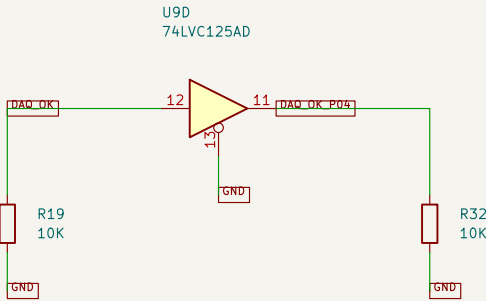
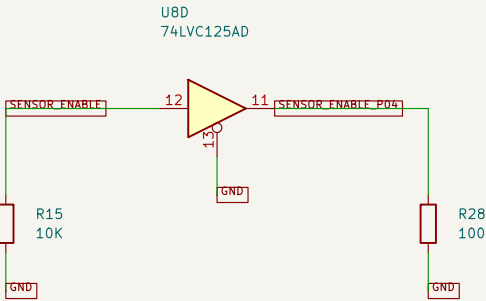
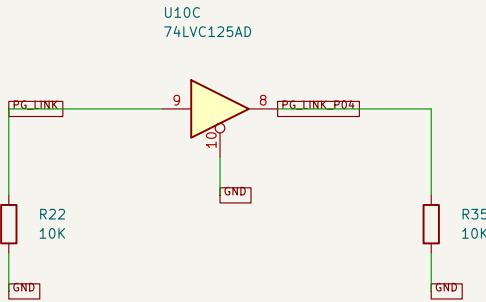
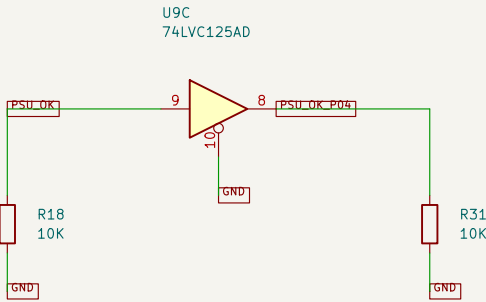
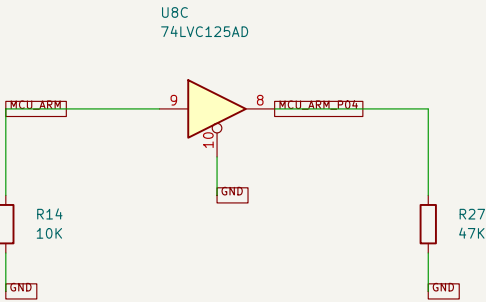
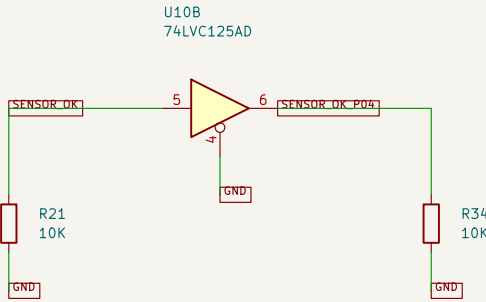
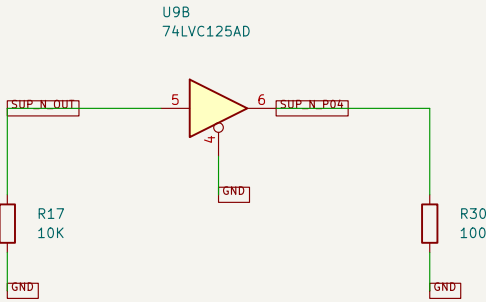
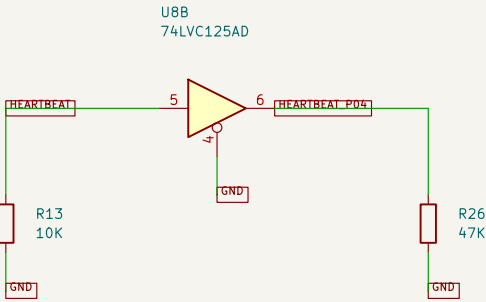
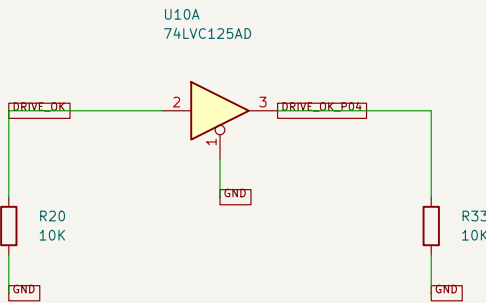
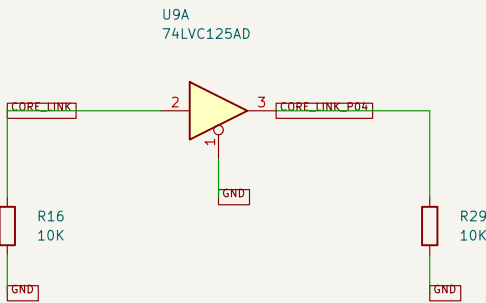
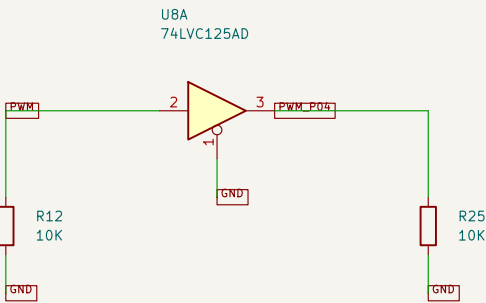
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Size: A3	Date: 2026-10-05
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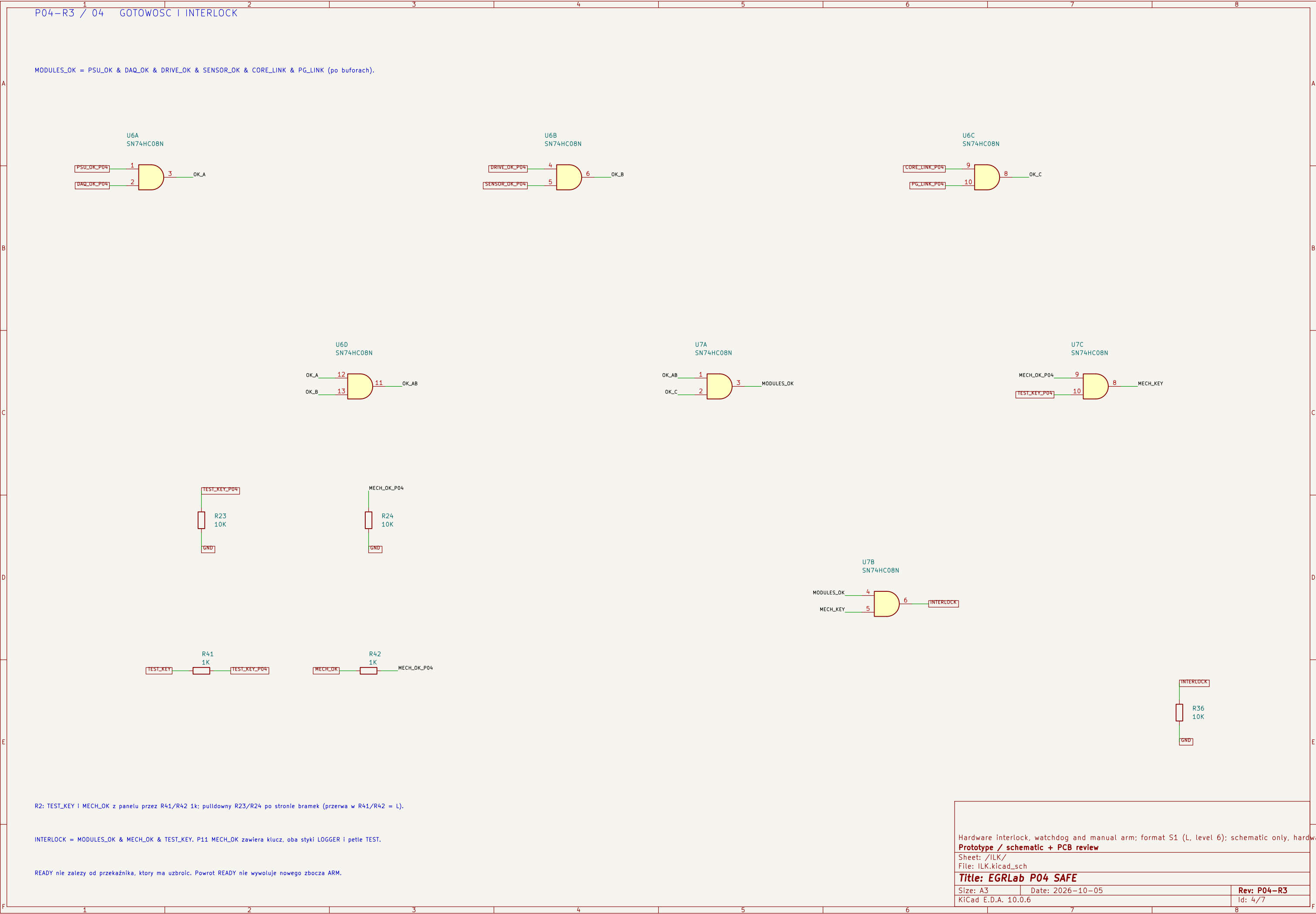
STEEP AS	
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74LVC125AD Nexperia: loff; pin wejścia nie zasila nieaktywnej domeny. Nie zastąpić układem HC125.

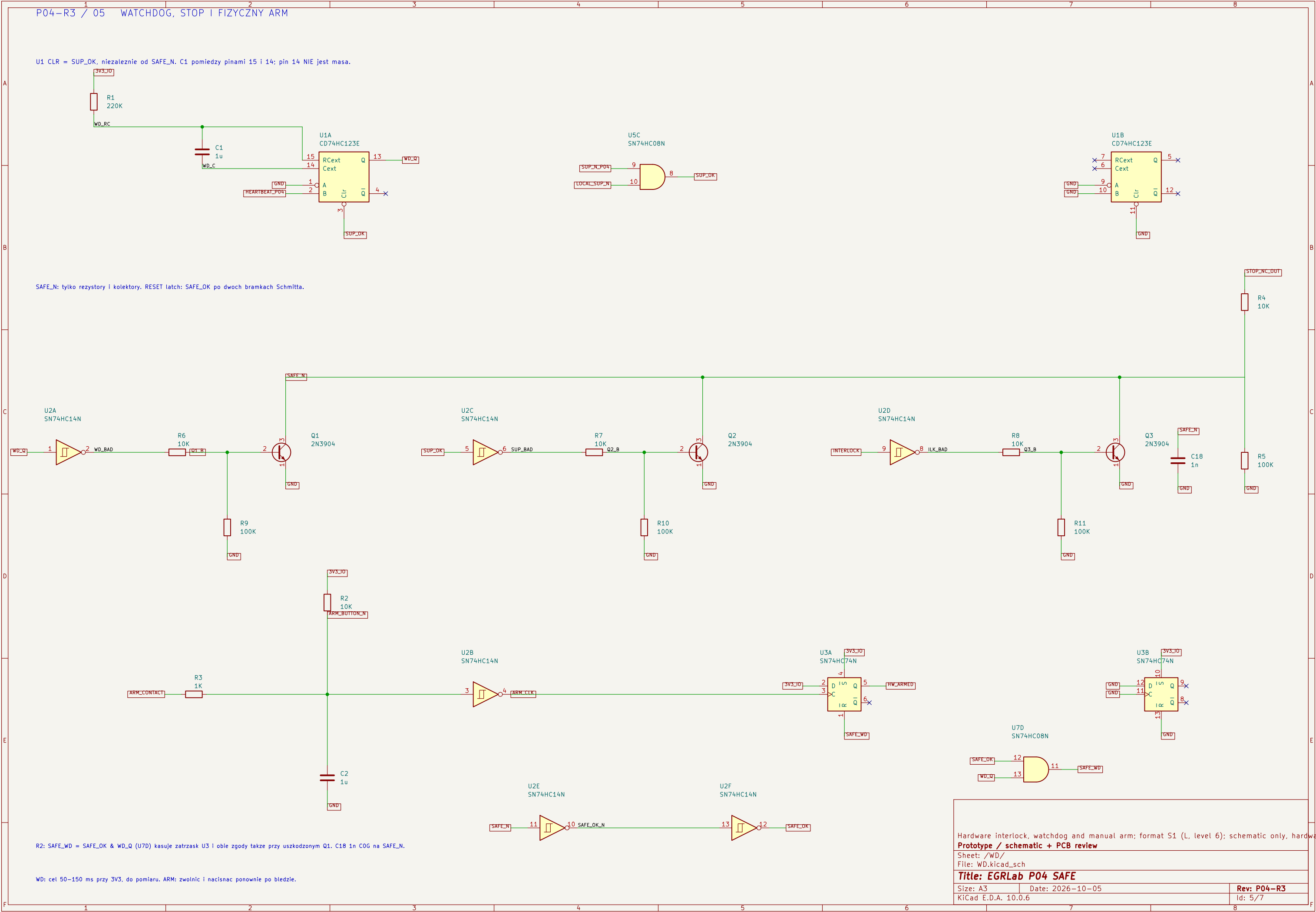


U1 CLR = SUP_OK, niezależnie od SAFE_N. C1 pomiędzy pinami 15 i 14; pin 14 NIE jest masa.

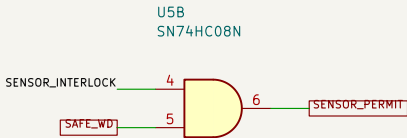
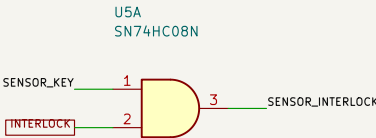
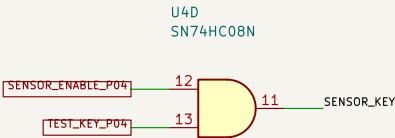
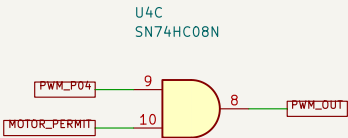
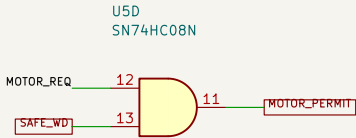
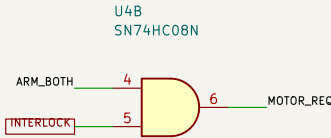
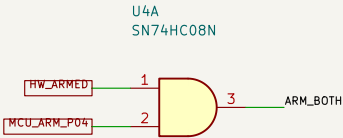
SAFE_N: tylko rezystory i kolektory. RESET latch: SAFE_OK po dwóch bramkach Schmitta.

R2: SAFE_WD = SAFE_OK & WD_Q (U7D) kasuje zatrzaśk U3 i obie zgody także przy uszkodzonym Q1. C18 1n COG na SAFE_N.

WD: cel 50-150 ms przy 3V3, do pomiaru. ARM: zwolnić i nacisnąć ponownie po błędzie.



MOTOR_PERMIT = HW_ARMED & MCU_ARM & INTERLOCK & SAFE_WD. PWM_OUT = PWM & MOTOR_PERMIT. SAFE_WD = SAFE_OK & WD_Q.



SENSOR_PERMIT = SENSOR_ENABLE & TEST_KEY & INTERLOCK & SAFE_WD. Fizyczny ARM nie jest potrzebny do samego czujnika.

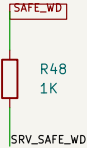
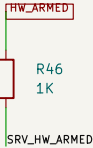
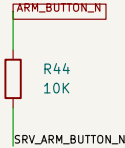
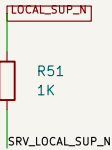
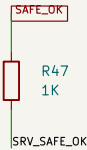
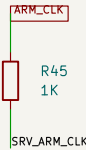
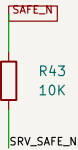
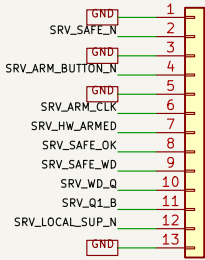
P07 pozostaje HOLD do kontroli rzeczywistego BTS7960. P04 nie ustala prądu ani kierunku mostka.

J_SV1 (S1, x=10..43): lancuch SAFE/ARM/watchdog. J_SV2 (S2, x=63.5..96.5): zgody. J_SV3 (S3, x=117..150): szyny. GND na koncach kazdej listwy.

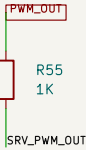
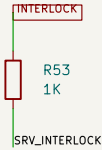
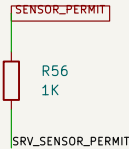
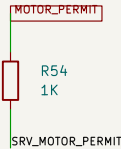
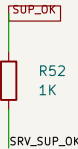
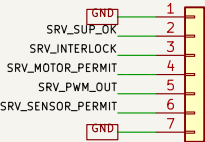
Rezystor przy wezle: 1k szyny i logika, 10k SAFE_N i ARM_BUTTON_N (wezly pasywne); Q1_B 1k (E21: zwarcie kolka z GND musi zat kac Q1).

SAFE_N i ARM_BUTTON_N tylko miedzy GND: mostek sondy z szyna nie moze podniesc SAFE_N przy otwartym STOP ani dac zbocza ARM.

J_SV1
J_SV1 SERWIS 1x13



J_SV2
J_SV2 SERWIS 1x7



J_SV3
J_SV3 SERWIS 1x7

